

Number Bond Fix-It Practice

Answer Key

Kindergarten–Grade 1

Quick Answers

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|------------------------------|--|
| 1. 5 | 4. the part that goes with $6 = 4$, — |
| 2. 3 | 5. $10 - 7 = 3$, — |
| 3. $5 + 4 = 9$, $9 - 5 = 4$ | 6. $2 + 8 = 10$, $10 - 2 = 8$, — |
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What is verified

3 of 6 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 5, 6. The worked solution states what a correct response must contain.

Curriculum

Level: Kindergarten–Grade 1

K.OA.A – *problems 1, 2, 3, 4, 5, 6*

Difficulty 1–4 of 5 across 11 tagged checks.

Common wrong answers

4. *If they got 5: kept the wrong part 5, checking it against itself instead of adding it to 6*
5. *If they got 7: copied the fact $10 - 3 = 7$ again instead of subtracting the other part 7*
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- 1.** Look at the number bond. The whole is 8. One part is 3. Write the missing part.
 $3 + \underline{\quad} = 8$

Solution: The two parts must add up to the whole, 8. One part is 3, so we count on from 3 up to 8: 4, 5, 6, 7, 8 — that is 5 counts. Check: $3 + 5 = 8$. The missing part is 5

- 2.** This bond is not finished. The whole is 10. One part is 7. Write the missing part.
 $7 + \underline{\quad} = 10$

Solution: The parts must make the whole, 10. Count on from 7 up to 10: 8, 9, 10 — that is 3 counts. Check: $7 + 3 = 10$. The missing part is 3

- 3.** The bond shows the whole 9 with parts 4 and 5. Two facts of its family are done: $4 + 5 = 9$ and $9 - 4 = 5$. Write the two missing facts.

Solution: A bond family uses the same three numbers in every fact. The two addition facts swap the parts, and the two subtraction facts take each part away from the whole.

- Addition with the parts swapped: $5 + 4$ is the same total as $4 + 5$, so $5 + 4 = \boxed{9}$
- Taking the part 5 from the whole: $9 - 5$ leaves the other part, so $9 - 5 = \boxed{4}$

4. Ana made this bond: the whole is 10, and she wrote the parts 6 and 5. Check her bond! Cross out the part that is wrong, and write the part that really goes with 6 to make 10. Then tell how you know.

Solution: Check Ana's parts: $6 + 5 = 11$, and 11 is too big — the whole is 10. So the part 5 is wrong and gets crossed out. Count on from 6 to 10: 7, 8, 9, 10 — that is 4 counts, and $6 + 4 = 10$. The correct part is $\boxed{4}$

Tell how you know (instructor judges): a good answer says that $6 + 5$ makes 11, not 10, so 5 cannot be a part; the part that goes with 6 must be 4 because $6 + 4 = 10$.

5. Sam wrote the fact family for the bond 10 with parts 3 and 7: $3 + 7 = 10$, $7 + 3 = 10$, $10 - 3 = 7$, $10 - 7 = 3$. One fact is written two times, so one fact is missing! Write the missing fact, and tell why Sam's list is not a full family.

Solution: Sam copied $10 - 3 = 7$ twice. The family still needs the fact that takes the other part, 7, away from the whole: $10 - 7$ leaves the part 3, so $10 - 7 = \boxed{3}$

Tell why (instructor judges): a good answer says a full family for one bond has four different facts — two additions and two subtractions — and Sam's list shows the same subtraction two times, so the family is missing $10 - 7 = 3$.

6. Challenge! Ben's bond shows the whole 10 with parts 8 and 2. He wrote only two facts: $8 + 2 = 10$ and $10 - 8 = 2$. Finish his family, and then tell why a bond with two different parts always has four facts.

Solution: Ben has one addition and one subtraction. The family still needs the addition with the parts swapped and the subtraction of the other part.

- Parts swapped: $2 + 8$ makes the same whole, so $2 + 8 = \boxed{10}$
- Take away the part 2: $10 - 2$ leaves the other part, so $10 - 2 = \boxed{8}$

Tell why (instructor judges): a good answer says the two parts can be added in either order (two addition facts), and each of the two parts can be taken from the whole (two subtraction facts) — with two different parts that makes four different facts.